Shakiba Shahidiani



AI Engineer | RAG & LLM Systems | Machine Learning | Evaluation

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Valid Dutch work permit holder

SUMMARY

AI Engineer with hands-on experience building end-to-end AI applications, from data ingestion and embedding-based retrieval to model evaluation and deployment. Combines a strong foundation in machine learning with practical experience developing RAG systems, multi-agent LLM systems and Python backend services. Turns ideas into reliable, measurable solutions by testing each approach against automated evaluation before it ships.

SKILLS

GenAI & LLM RAG, LLM Evaluation (LLM-as-Judge), Guardrails, Multi-Agent Systems, Prompt Engineering, Chunking Strategies, Embeddings, Vector Search (ChromaDB, Pinecone, FAISS), Structured Outputs, LangChain, LangGraph, LiteLLM, Fine-Tuning (LLaMA, BART)
Machine Learning PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, CNNs, Transfer Learning, GNNs, LSTMs, Knowledge Graphs, Feature Engineering, Statistical Validation, Regression Analysis, A/B Testing
Engineering Python, FastAPI, SQL, PostgreSQL, SQLAlchemy, Pandas, pytest, Docker, Git, GitHub Actions CI/CD, AWS (EC2, RDS, VPC), Terraform, Streamlit, Claude Code, Codex

EXPERIENCE

Corteva Agriscience **Sep 2024 – Sep 2025**
Data Scientist *Italy*

- Replaced a manual Excel forecasting workflow with an end-to-end predictive pipeline on ten years of seed production data, giving planners a repeatable, data-driven forecast of seed longevity.
- Reconstructed unified batch histories by resolving inconsistent ID mappings across multi-stage production pipelines, making complete batch trajectories available for model training.
- Engineered features from seed breed, morphology and processing history and trained XGBoost and Random Forest models, validated against the legacy approach.
- Worked with seed technology specialists to turn domain knowledge into features and data requirements, and presented results, confidence and limitations to management at each stage.

Master's Thesis, Graph ML for Drug-Gene Interaction Prediction **Mar 2025 – Oct 2025**
Politecnico di Milano *Italy*

- Built a heterogeneous biomedical knowledge graph of 45,000+ nodes and 700,000+ edges integrating drugs, genes, pathways and Gene Ontology terms for drug repurposing via link prediction.
- Learned graph embeddings with metapath2vec and domain-informed metapaths, reaching 97% top-10 recall (Cohen's d = 0.93 over baselines) under a 5-seed framework with multi-scenario leave-out validation to prevent leakage.

PROJECTS

RAG Assistant for Enterprise Policy Documents **Jun 2026**

- Built an async FastAPI RAG service over policy PDFs with section-aware chunking, ChromaDB vector search and a chat interface returning answers with inline source citations.
- Designed a metadata-driven version-resolution layer that hard-filters superseded documents at query time, so answers cite only the policy currently in force.
- Built an automated evaluation harness on a labelled golden set scoring retrieval recall, fact coverage, answer faithfulness (LLM-as-judge) and guardrail accuracy; it caught two regressions that looked like improvements before measurement. Grounding guardrails make the system abstain when no relevant document is found.

LLM Debate System, Multi-Agent AI **Apr 2026**

- Built a multi-agent system using LiteLLM to orchestrate persona-based LLM agents across debate rounds, with a moderator agent that evaluates consensus and conditionally ends the debate, with explicit state transitions, deadlock fallback and PostgreSQL persistence.
- Deployed the containerised application on AWS (EC2, RDS, VPC) using Terraform infrastructure as code and a GitHub Actions CI/CD pipeline for automated deployment.

Automated Long-Form Question Answering and Content Verification **Mar 2024**

- Fine-tuned LLaMA and BART in a RAG pipeline, comparing retrieval strategies and prompt formats with ROUGE, and built a classifier separating human-written from AI-generated text.

Plant Health Diagnostics via Computer Vision **Sep 2023**

- Built a CNN image classification pipeline with transfer learning and augmentation for a 38% positive class, an out-of-distribution filter that removed artifacts and label noise, and self-supervised pre-training (MoCo) on unlabelled images.

EDUCATION

MSc Computer Science and Engineering, Politecnico di Milano, Italy Sep 2022 – Oct 2025
BSc Software Engineering, University of Science and Culture, Tehran Sep 2016 – Apr 2021

LANGUAGES

English (C1, professional working proficiency) | Persian (native)